

A learners' microcomputer

With this installment we commence construction details of the microcomputer. To ensure success, it is built step by step and checked at each stage.

Design: Hugh Anderson
Development: Graeme Teesdale

Part 3

BEFORE commencing construction, no doubt just about every constructor is going to ask "what will it cost?". The answer is, "that depends"! This project can be built up in a number of forms. You can elect to construct it with only the minimum amount of memory (RAM), 1K, and monochrome video output. This saves some component costs. If you have a suitable power transformer or plug pack, then you won't need to buy one, saving something there. A standard hex keypad can be used, rather than the one we have designed which uses individual pushbuttons, and this saves a few dollars too. Rather than a metal case with wooden ends, etc, the project can be mounted on a piece of chipboard, thus saving the cost of the case. You can also make your own pc board and save on that too. It all depends on how resourceful you are. If you shop around for the best prices in components, particularly the ICs, you can save a considerable amount. If you're game, or experienced, the IC sockets can be dispensed with, too (except for the ROM, IC11, but you'll see why later).

Let us suppose a 'minimum' version would consist of all the components required to produce monochrome video, a ready-made pc board, the ETI-760 Video RF Modulator kit and a suitable plug pack or low voltage transformer. With a little shopping around you can obtain it all for a total price that ranges from around \$98 to \$105. Without attempting to get best prices, it may cost up to \$130. Case hardware would cost extra. The case to be described is quite simple and can be built with a minimum of tools, but we'll get to that later.

The 'complete' project, which shall we say includes the colour expansion com-

ponents and full 3K of RAM, IC sockets and case, we calculate may cost from \$130 up to \$180. It is obvious that the variation in component prices can be enormous, and this accounts for the wide variation. For example, the extra four 2114 RAM ICs to provide a memory capacity of 3K on-board may cost as low as \$8 or as much as \$19.80. A set of IC sockets may cost as little as \$3 or as

much as \$9. We would expect kit suppliers to offer kits in differing versions so it's a little hard to predict what a kit may cost you.

The board

The pc board has been designed so that a constructor with experience in producing his own at home can do so. It is a double-sided board and connections

FEATURES

- Low cost
- Simple construction
- Widely available components
- Both colour and B&W video
- RF modulator permits connection to any TV set
- Powered from a plug pack (battery possible too!)
- 3K of on-board memory (expandable to 64K off-board)
- Cassette interface on-board
- Audio output (tones programmable)
- Hex keyboard for simple programming, games playing
- Audible tone indicates key being depressed
- Additional 8-bit input/output port
- CHIP-8 monitor program
- User socket for prototyping, additions, etc.
- Learn as you build
- Teach yourself programming (articles to follow)

EXPANSION

- Standard ASCII keyboard
 - Additional RAM
 - Light pen
 - Sound effects
 - + more!
-

from the top side to the bottom side are made with feedthrough links. All the pads for the IC pins are on the bottom side of the board so that no IC pins have to be soldered both sides. You will notice that these pads are quite small as tracks pass between them on the board. Take care not to over-etch the board otherwise you may have tracks disappearing!

Our pc board design includes a keyboard which is made up using individual pushbuttons (Fujisoku, type FES 310 from IRH Components). If you elect to use a standard hex keypad then this section of the board is unnecessary. Dotted lines adjacent to the 8 Vac input and the location of R2 and R3 indicate where the board finishes if you dispense with our keyboard.

If you're making your own board, the next step after etching is a visual inspection for bridges between tracks and breaks in tracks. Having done that and fixed up any problems, the board can be drilled. This has to be done in stages. First, drill *all* holes with a No. 73 drill. The pads for the IC pins are quite small, to allow tracks to run between them, hence the small size drill bit. Drill the holes from the *bottom side* of the board. It is important to drill accurately.

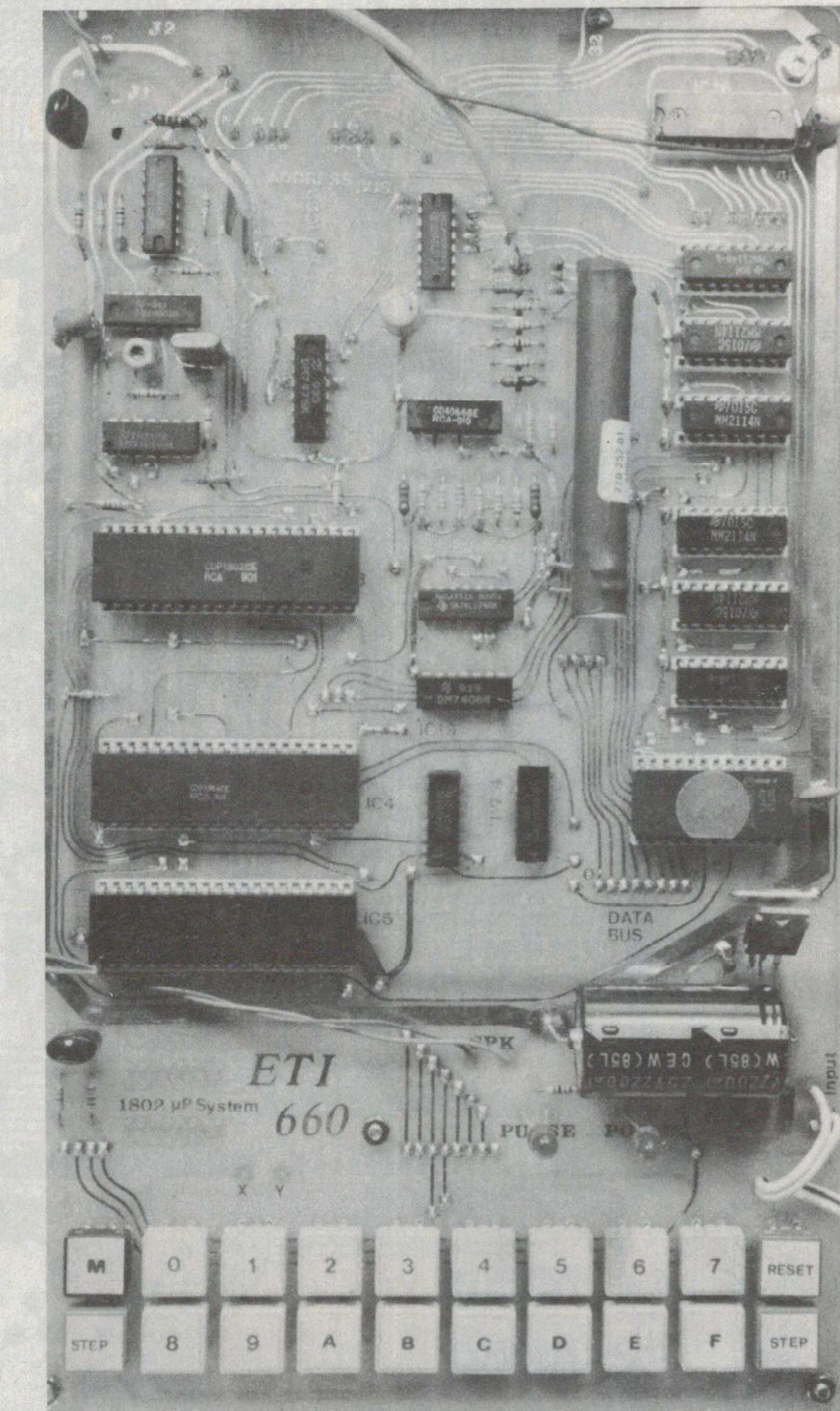
When you think you have finished this part of the drilling, hold the board up to the light — and see which holes you've missed! Having completed this stage, another visual inspection is necessary. Carefully examine *both sides*

ETI-660 PCB ARTWORK

NOTE: The pc board artwork is too large to reproduce in the magazine, unfortunately. However, home constructors intending to make their own boards may obtain prints by sending a large (at least 300 x 250 mm) stamped, self-addressed envelope to ETI-660 PCB, ETI Magazine, 15 Boundary St, Rushcutters Bay NSW 2011.

of the board and see that no tracks have been cut through by the drill. Mark any suspects with a chinagraph pencil or a dab of paint, for fixing later.

The next stage is to drill holes to take component leads and sundry other holes. To identify these holes we have provided an overlay drawing. Feed-through connections between the top and bottom of the board are also shown on this diagram, but we'll get to them shortly. On this diagram, holes marked with a circle are to be drilled to take component leads. Use a No. 60 drill bit for these. Where the component pad is



The fully assembled board, less IC22 and IC23, which are only necessary with later expansions. The long cylindrical object at upper right is the 400 ns colour delay line. The IC at middle right is the EPROM containing the CHIP-8 monitor program.

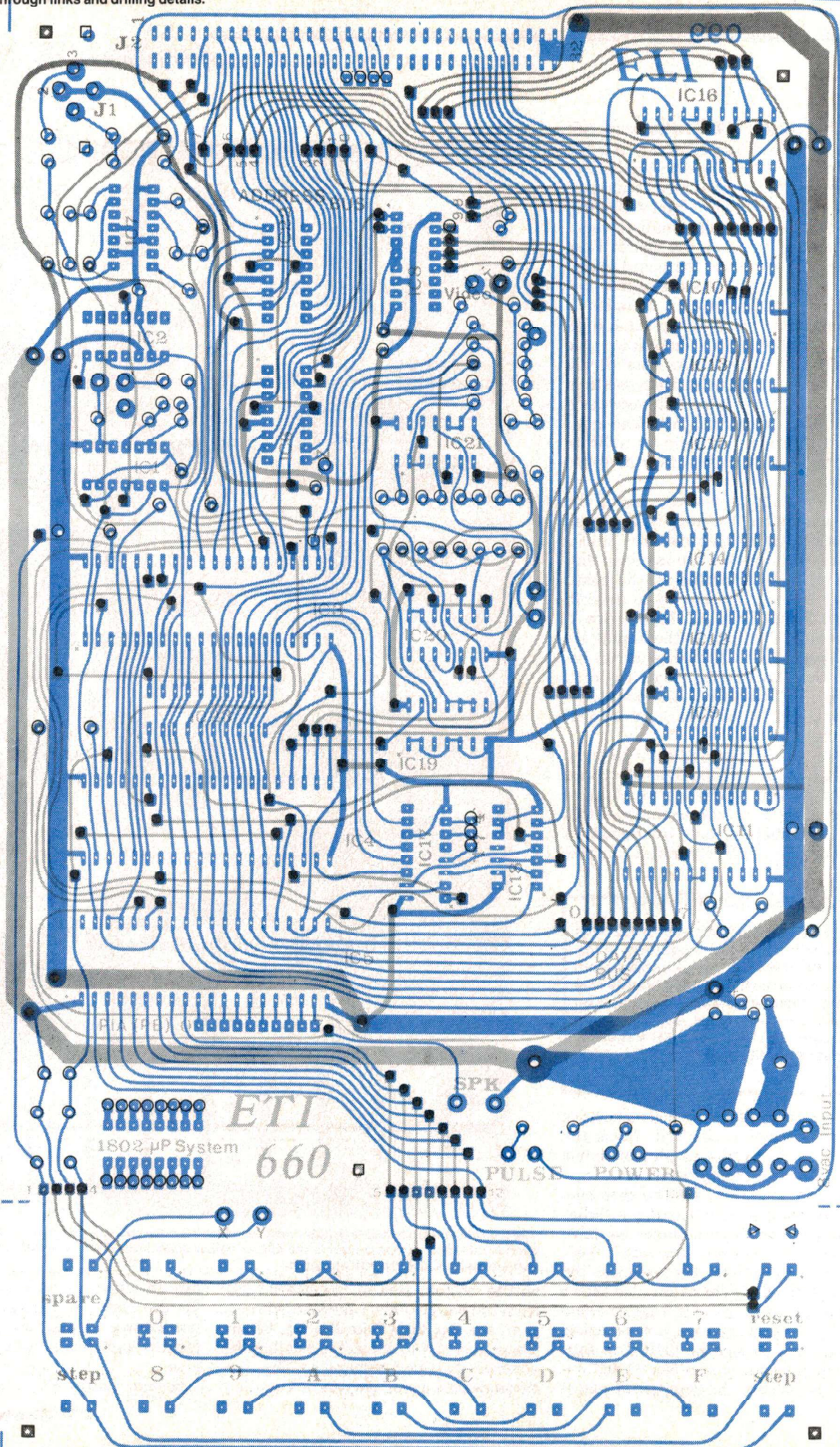
on the bottom side of the board, drill from the bottom side. Likewise, where it is on the top, drill from the top. Where there is a pad to a track on both sides you may drill from either side. The feed-through link holes may be left as they

are, in which case 26 swg wire is used for the links (see later). If you are to use 22 swg wire for the links, drill the feedthrough holes with the No. 60 bit also.

Seven holes are marked with a

● FEEDTHROUGH LINK HOLES, DRILL EITHER NO. 60 OR NO. 73, 188 HOLES
ALL OTHER HOLES DRILLED NO. 73

□ MOUNTING HOLE, DRILL 3 mm DIA., 7 HOLES
△ DRILL 4 mm DIA., 2 HOLES
○ COMPONENT LEAD HOLES, DRILL NO. 60, 153 HOLES



square. These are for mounting bolts and should be drilled to 6 BA clearance (say, 3 mm). Note that the cassette interface socket, J1, can be mounted on-board and two bolt holes are provided. Two holes adjacent to the 8 Vac input are marked with triangles. These should be drilled to 4 mm diameter and serve to secure the input lead from the plug pack.

Having completed the drilling, once again make a visual inspection to see that nothing has gone amiss and check that you have drilled all the necessary holes.

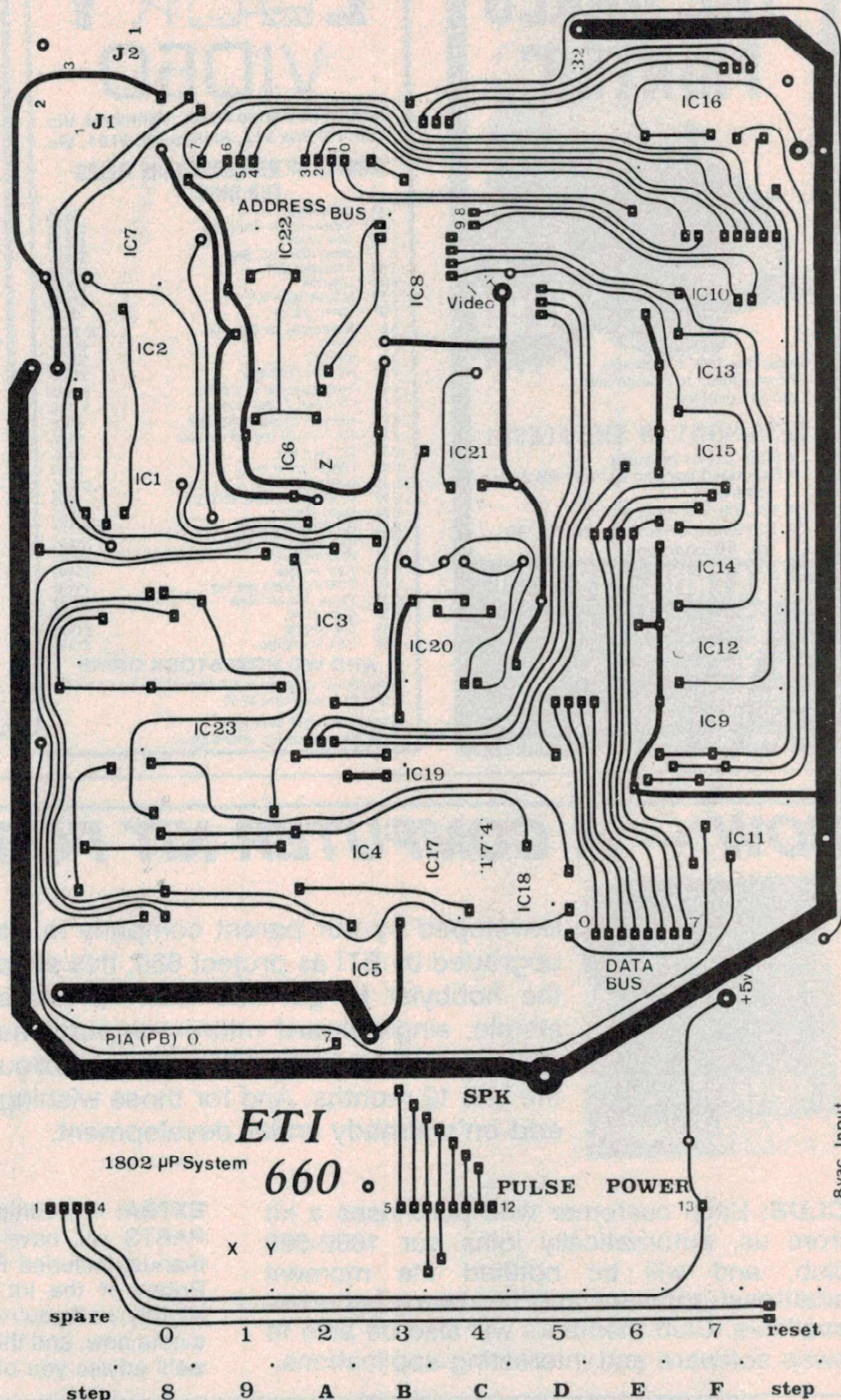
Even if you have purchased a ready-made pc board it is a good idea to make a thorough visual inspection of the board to ensure that there are no cut tracks and no bridges between tracks. It will save hassles later.

You will note that we have provided some annotation on the pc board. All the IC positions are identified. Pin 1 of each IC is marked with an adjacent 'spot' on the top side of the board. The data buss and address buss lines are identified in convenient positions. The cassette interface socket, J1, and the expansion socket, J2, are both identified (even though they may be mounted off-board). The video output connection is identified as is the +5 V rail at the output of the three-terminal regulator. The 'power' indicator (LED1) and the 'pulse' indicator (LED2) are both indicated as well as the speaker connections (SPK). At the keyboard end of the board you'll find a number of pads in a row marked '1...4', '5...12' and '13'. These are connections for an external hex keypad and 'step' and 'reset' pushbuttons. All the keys for our on-board hex keyboard are identified. Connections to the 'spare' key are designated 'X' and 'Y'. Adjacent to IC6 is a terminal marked 'Z'. This is an output signal from the 1802 micro-processor and you don't have to worry about that for some time to come. Between IC17 and IC18 are three terminals marked '1', '7' and '4'. These are decoded outputs from the 1802 and again, you don't have to worry about them. The 8-bit 'user port' (I/O port from the PIA, IC5) is identified, adjacent to IC5.

Have a good look at the pc board and familiarise yourself with all the annotations so you can 'find your way around' as you construct the project.

Feedthrough links

The next stage of construction is to install all the feedthrough links. The board was designed this way so that the IC pins only need to be soldered on the bottom side of the board (non-



Reproduction of the topside of the pc board (not to scale) showing the annotation included to aid assembly and major component identification.

component side). This makes the ICs easy to remove later if you get one in the wrong way round or if a fault necessitates later removal. We recommend you fit sockets for about half the ICs in any

case, but we'll get to that shortly.

Obtain two strips of wood or plastic about three of four millimetres thick (two rulers are ideal!). Place them on the bench or table about 250-260 mm

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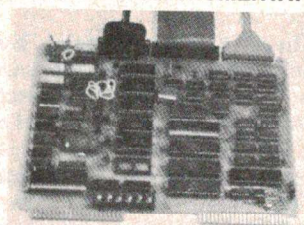
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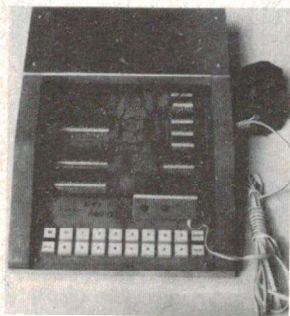
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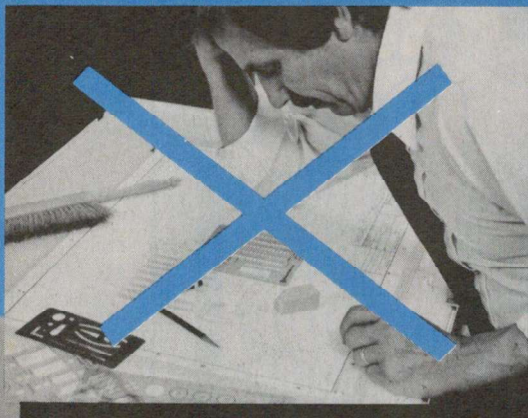
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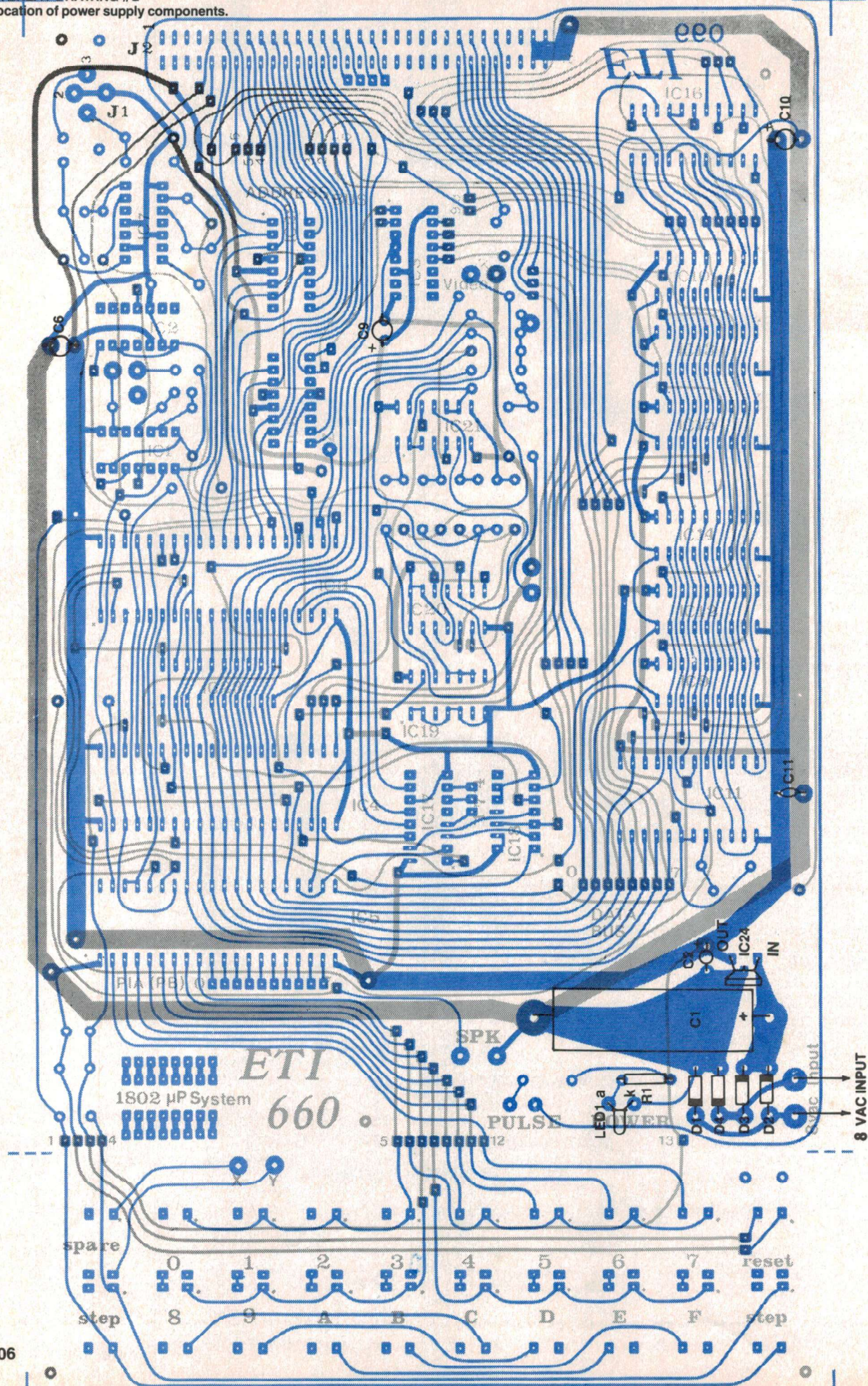
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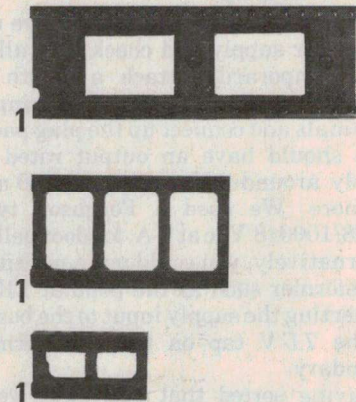
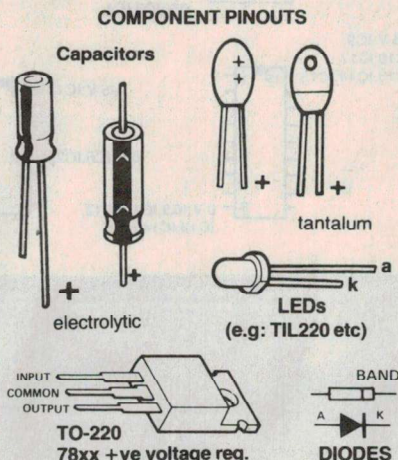
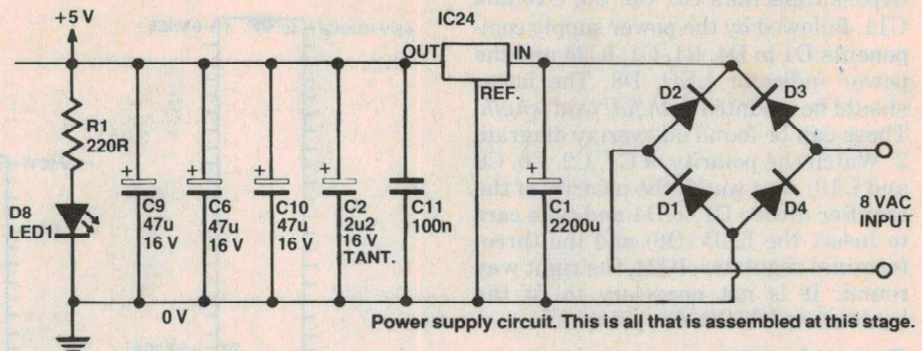


apart and lay the pc board on them, component side up (i.e: annotations up) For linking wire we used 26 swg (0.455 mm dia.) tinned copper wire. If you find 22 swg tinned copper wire easier to obtain then all the linking holes should be drilled with a No. 60 bit too, as for the component lead holes.

Note that, on the component side of the board, the track pads for the links are all square types. Component lead pads are round.

Referring to overlay drawing 1, which details the feedthrough links, insert the wire into each hole, passing it through until it stops against the bench or table top, solder it to the pad and then cut the wire. When you think you have done all the links, check carefully against the drawing — to find the ones you've missed! Now you can turn the board over and solder the links to the pads on the bottom side. Trim off the excess wire. Solder quickly and cleanly with a clean, hot iron with a small radius tip and you should have little difficulty. Turn the board over and check the top side to see that no joints have been disturbed or become frosty in appearance, indicating a possible dry joint. Check that those links *under* the ICs are installed. Once you have an IC socket or an IC in place, you'll find it very difficult to fit any missed links underneath them!

If you are going to use IC sockets, install them now. Even if you're not, we recommend you install an IC socket for IC11 (the EPROM). We fitted sockets for IC3, IC4, IC5 and IC9 to IC16 inclusive.



IC sockets showing how to identify pin 1.

Make sure you fit each socket the right way round and that no pins get folded under the socket. The accompanying picture shows how to identify pin 1 on a socket. Remember that on the pc board pin 1 on each IC position is identified by an adjacent spot on the top side of the board.

Take care when soldering to IC pads that have conductors running between them. It is easy to get a solder 'bridge' across the tracks that will later cause the project to malfunction. A visual check after soldering and/or using an ohmmeter to check that no bridging has occurred is recommended.

COMPLETE PARTS LIST ETI-660 LEARNERS' MICROCOMPUTER

Parts marked with * are only used when colour expansion is included. Parts marked ** change when colour expansion added.

Resistors

R1	220R
R2,R27,R30	120k
R3	47k
*R4,R6,R19	4k7
*R5	1k
**R7	3k3 (2k2 for colour)
*R8	680R
*R9	68R
*R10	1k2 (470R for colour)
*R11,R16,R17	2k2
**R12	10k (5k6 for colour)
*R13,R18	1k2
*R14	1k5
*R15	2k7
R20,R21,R22	390R
R23	15k
R24	1k2
R25	8k2
R26	180R
R28,R32	22k
R29	10k (only for high-order EPROM, see text)
R31,R33	1M
R34	180k

Capacitors

C1	2200u/16 V axial electro.
C2	2u2/16 V tant. or RBLL
C3,C8,C11	100n greencap
C4	10n greencap or ceramic
C5	3-30p film or ceramic trimmer
C6,C9,C10	47u/16 V tant. or RBLL
C7	470p ceramic

Semiconductors

D1-D4	1N4001, EM401 or sim.
D5,D6	1N914, 1N4148 (only for higher order EPROM, see text)
D7	(LED2) TIL220R or sim. red LED
D8	(LED1) TIL220G or sim. green LED
IC1	74LS00
IC2	7490
IC3	CDP 1802
IC4	CDP1864
IC5	MM6821
IC6	7475
IC7	LM3900
IC8	74LS138
IC9,IC10	2114
IC11	2516, 2716 EPROM (with CHIP8)

IC12-IC15	2114 (optional, for RAM expansion)
*IC16	MM2101, 5101, CDP 1822 etc.

IC17	4028
*IC18	74LS00
*IC19	74LS68
*IC20	74LS74
*IC21	4066

NOTE: IC22, IC23 are only used with subsequent additions to this project and while provision has been made for them on-board they are not used at this stage.

IC24	LM340T/5, 7805 +ve 5 V reg.
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Miscellaneous

*DLY	400 ns colour delay line, Plessey 220-252-01
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ETI-660 pc board; 3-pin 180° DIN socket; 3-pin 180° DIN plug; 18 type FES-310 pushbuttons or standard HEX keypad; 50 or 75 mm diameter 8 or 16 ohm speaker; 8.867238 MHz crystal; three 40-pin IC sockets; six 18-pin IC sockets; one 22-pin IC socket; one 24-pin IC socket; Ferguson plug pack type PPB 8/1000 or similar (8 Vac @ 1 A output); aluminium top and bottom panels as per drawings; heatsink for IC24 as per drawing; wooden end cheeks as per drawing; mounting hardware, etc. (Mechanical drawings next month.)

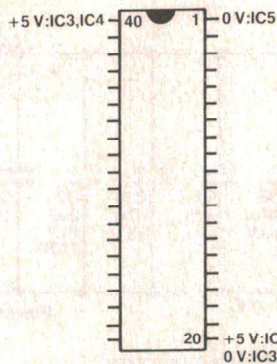
Next step is to fit the supply rail bypass capacitors C2, C6, C9, C10 and C11, followed by the power supply components D1 to D4, R1, C1, IC24 and the power indicator LED, D8. The latter should be mounted *with full lead length*. These can be found on overlay diagram 2. Watch the polarity of C1, C2, C6, C9 and C10. Also watch the polarity of the rectifier diodes D1 to D4 and take care to insert the LED (D8) and the three-terminal regulator, IC24, the right way round. It is not necessary to fit the heatsink to IC24 at this point.

Powering up

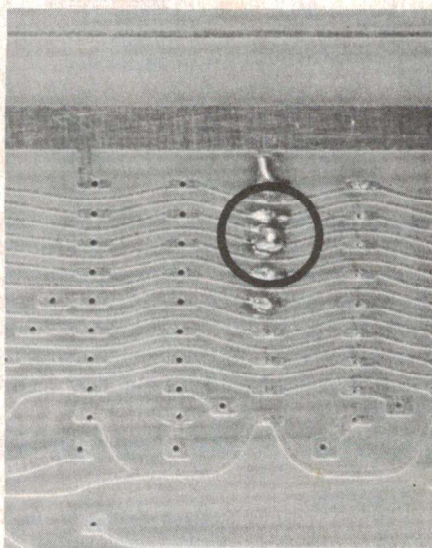
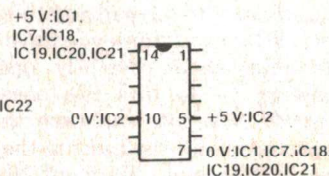
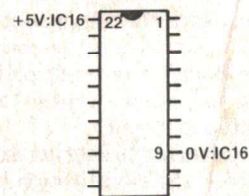
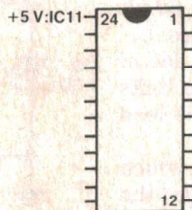
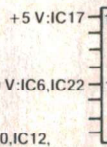
At this stage you're ready to 'fire up' the power supply and check that all is OK. Temporarily attach a length of figure-eight flex to the 8 Vac input terminals and connect up the plug pack. This should have an output rated to supply around 8 Vac at up to 500 mA or more. We used a Ferguson type PPB8/1000 (8 Vac at 1 A, for door bells). Alternatively, you could use a multitap transformer such as the popular 2155, connecting the supply input to the board to the 7.5 V tap on the transformer secondary.

Having sorted that out, place your trusty multimeter nearby and apply power. Check that there are no wisps of smoke rising from anything. The power indicator LED should be on. If it's not, switch off and check that it's inserted the right way round. If it is, check that the rectifier diodes are inserted the right way round, along with IC24. If they seem OK, then something else is likely to be wrong. To check, remove IC24 and apply power once again. Measure the voltage across C1 (negative lead of multimeter to negative lead of capacitor, positive lead of multimeter to positive lead of capacitor). You should get around 12-14 V. If you don't, you've got one or more of the rectifier diodes inserted incorrectly. Switch off and correct it if necessary. If all is OK, switch off and check that you haven't got a short somewhere around the supply rail tracks on the pc board. Turn the board over. Take your multimeter and measure the voltage across the supply pins of each IC socket or position on the board. You should get 5 V connecting your meter as follows:

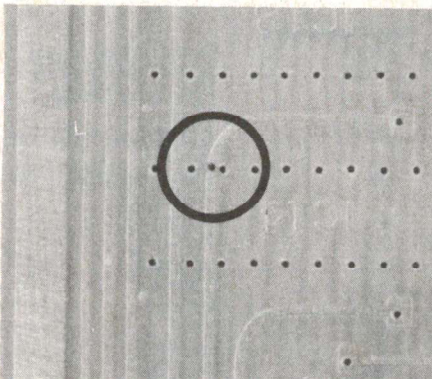
- IC1, IC7, IC18, IC19, IC20, IC21 pin 7 (negative lead) and pin 14 (positive lead).
- IC17 pin 8 (negative lead) and pin 16 (positive lead).
- IC6, IC22 pin 12 (negative lead) and pin 5 (positive lead).
- IC2 pin 10 (negative lead) and pin 5 (positive lead).
- IC3 and IC4, pin 20 (negative lead) and pin 40 (positive lead).



+5 V:IC9, IC10,IC12, IC13,IC14,IC15



Watch for solder 'bridges' across adjacent tracks when you are soldering IC pins or IC socket pins. This is particularly important around IC3, IC4, IC5 and ICs 9 to 16. Remove any bridges that you find with a solder sucker or solder wick, then carefully re-solder the pins.



Take care when drilling the board that your aim is not off and the drill cuts a track, as shown here. A careful visual examination of the board will show up such problems.

UNDERSIDE VIEW OF ICs showing where power supply pins are located.

- IC5 pin 1 (negative lead) and pin 20 (positive lead).
- IC9, IC10, IC12, IC13, IC14, IC15 pin 9 (negative lead) and pin 18 (positive lead).
- IC11 pin 12 (negative lead) and pin 24 (positive lead).
- IC16 pins 8/9 (negative lead) and pin 22 (positive lead).

Checking at the pins of each IC location like this eliminates ground (0 V rail) return errors.

Next to check are the data and address buss lines. Connect the negative probe of your multimeter to the 0 V (ground) rail — that's the wide track that runs around the top of the board from the 'pin 32' end of J2 to near IC2. Now, with the positive probe of your multimeter, check that there is *no voltage* on each of the following:

- Address buss lines 0 to 9
- Data buss lines 0 to 7

The address buss lines are identified near IC22 and the data buss lines are identified adjacent to IC18 and IC11, on the top of the board.

This test should be repeated in the following way: connect the multimeter positive probe to the +5 V rail (see the pad adjacent to IC24 — it says "+5 V") and check that no voltage appears on the address or data buss lines using the negative probe of the multimeter.

If you have any faults, trace them and clear them now. Most likely you will find them to be solder bridges between tracks at IC socket pins, but you may find small 'fingers' bridging tracks where the copper has not been etched away. This is something to watch if you have made your own board.

If all checks out well, you're ready to go on with the next stage.

(Continued next month).